Sleep Under The Sea!

Exploring the evolutionary biology of sleep architecture through marine mammals

Kirya Caine

Learning objectives

1. Understand traditional and modern theories of sleep evolution
2. Explore how marine mammals present a unique model to explore how ecology influences sleep
3. Appreciate why the question of the evolutionary basis of sleep is so interesting to explore and difficult to answer!

Reason for pursuing topic

- Background in evolutionary biology
- NSCI 251 centered around disorders/health concerns associated with sleep perturbations
- But in the animal world, even among mammals, huge variance in sleep duration and architecture







Previous/intuitive evolutionary models of sleep

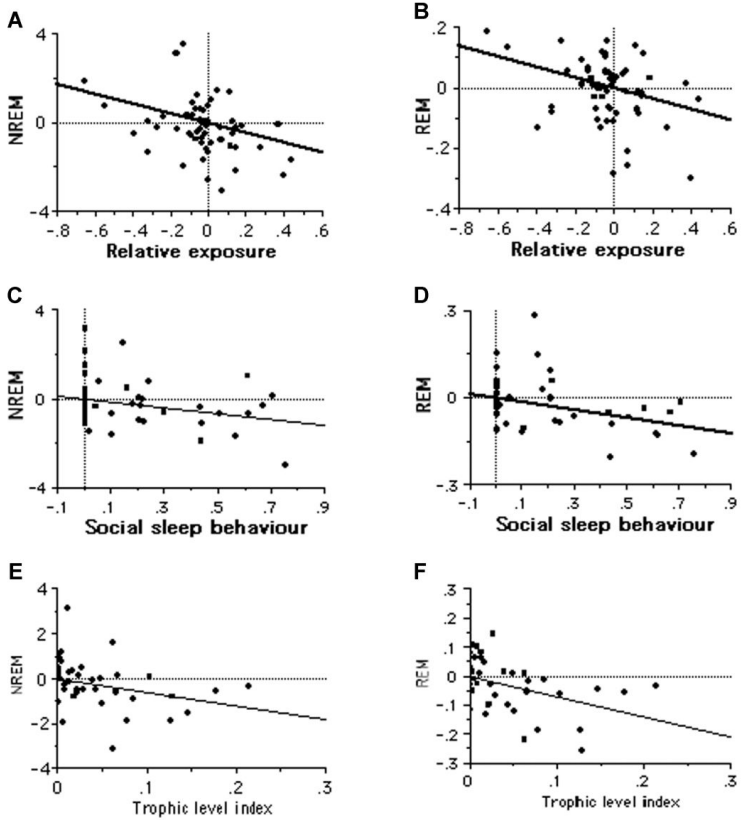
- Historical theories of sleep and evolution based on observations in humans and laboratory animals
    - Increased sleep duration must be an evolutionary building block required for encephalization
    - Increased sleep duration allows for a higher basal metabolic rate

Capellini, I., Barton, R. A., McNamara, P., Preston, B. T., & Nunn, C. L. (2008). Phylogenetic analysis of the ecology and evolution of mammalian sleep. Evolution; international journal of organic evolution, 62(7), 1764–1776. https://doi.org/10.1111/j.1558-5646.2008.00392.x

Previous/intuitive evolutionary models of sleep... are **WRONG**

- Comparative sleep patterns do not actually significantly correlate with functional features hypothesized to drive them: relative brain size, BMR, or brain development.

- Larger role than expected of predation, foraging, and other ecological factors on the duration of both NREM and REM sleep.



Capellini, I., Barton, R. A., McNamara, P., Preston, B. T., & Nunn, C. L. (2008). Phylogenetic analysis of the ecology and evolution of mammalian sleep. Evolution; international journal of organic evolution, 62(7), 1764–1776. https://doi.org/10.1111/j.1558-5646.2008.00392.x

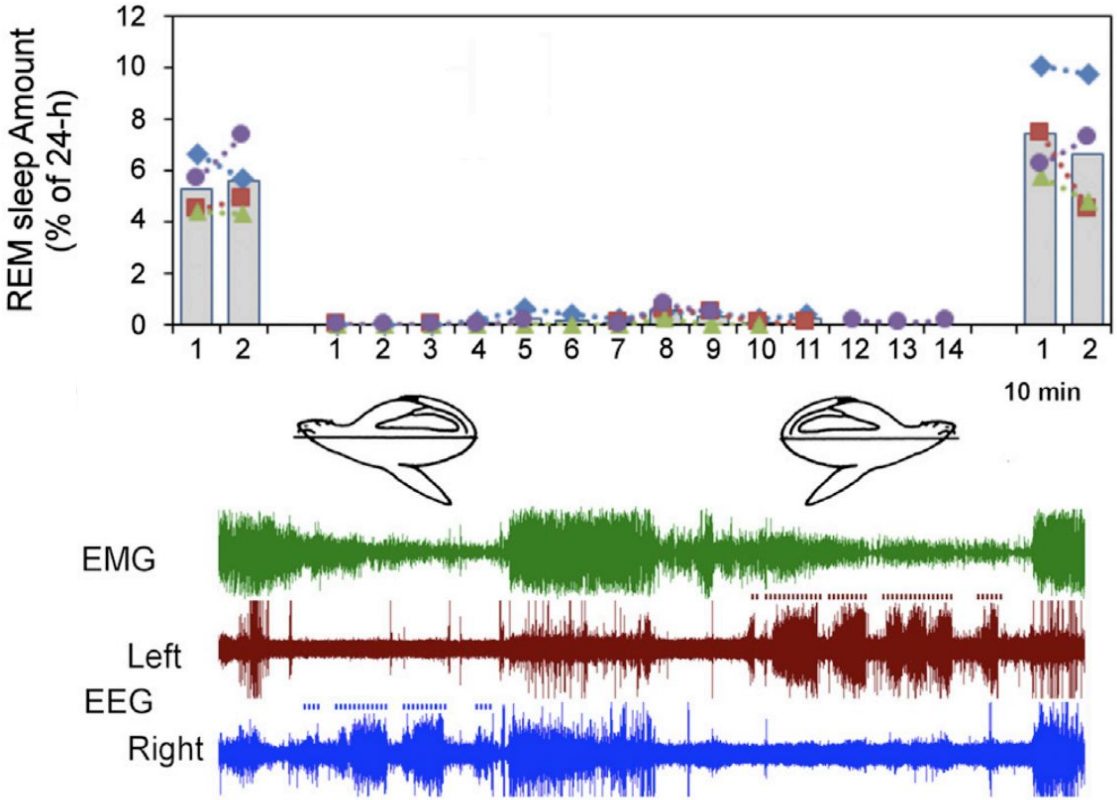
Testing new theories of sleep

- Marine mammals provide a great model for studying effect of ecology on sleep
    - High ecological costs of sleep in the ocean



Fur seals dont catch up on REM sleep, exhibit USWS

- Only exhibit bilateral slow wave sleep (BSWS) and REM sleep when resting on land.
- Exhibit unilateral slow wave sleep (USWS) at sea, where where only one hemisphere of the brain exhibits sleep-like wave patterns at a time.
    - Don’t appear to experience REM rebound after



Lyamin, O. I., Kosenko, P. O., Korneva, S. M., Vyssotski, A. L., Mukhametov, L. M., & Siegel, J. M. (2018). Fur Seals Suppress REM Sleep for Very Long Periods without Subsequent Rebound. Current biology : CB, 28(12), 2000–2005.e2. https://doi.org/10.1016/j.cub.2018.05.022

Cetacean USWS

- USWS makes up 70–90% sleep in cetaceans with little/insufficient evidence of REM
    - Asymmetrical eye closure
    - BSWS appears to interfere with breathing



Lyamin, O. I., & Siegel, J. M. (2019). Sleep in Aquatic Mammals. Handbook of behavioral neuroscience, 30, 375–393. https://doi.org/10.1016/b978-0-12-813743-7.00025-6

Cetacean USWS

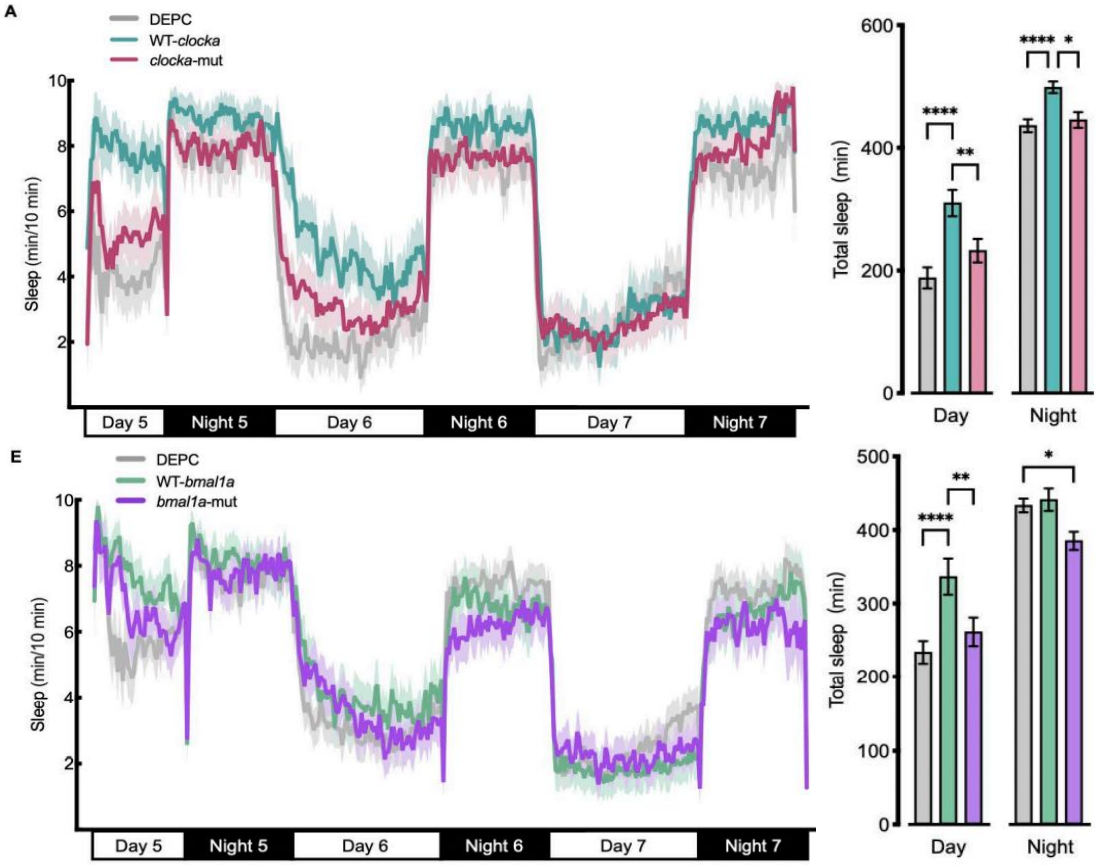
- Sleep features show signatures of ecological pressures
    - “Resting” near-surface sleep disappears for weeks postpartum, mothers and calves constantly in motion → regulation of predation, body temperature
    - Dolphins can continuously generate echolocation clicks when sleeping



Lyamin, O. I., & Siegel, J. M. (2019). Sleep in Aquatic Mammals. Handbook of behavioral neuroscience, 30, 375–393. https://doi.org/10.1016/b978-0-12-813743-7.00025-6

Genetic signatures of sleep evolution in cetaceans

- Convergent, positive selection on sleep related genes
  - Including CLOCK and BMAL1
- Zebrafish studies of functional consequences:
  - gene expression: downregulation of sleep-promoting pathways, upregulation of glutamatergic transmission
  - Less sleep overall



Yin, D., Zhong, Z., Zeng, F., Xu, Z., Li, J., Ren, W., Yang, G., Wang, H., & Xu, S. (2025). Evolution of canonical circadian clock genes underlies unique sleep strategies of marine mammals for secondary aquatic adaptation. PLoS genetics, 21(3), e1011598. https://doi.org/10.1371/journal.pgen.1011598

Key points

- Historical evolutionary theories of sleep that rely solely on the functional features of sleep are inadequate
- Comparative studies indicate a greater contribution of ecological sleep-associated costs than previously thought
- Marine mammals → interesting case study demonstrating how changes in sleep architecture are correlated with changes in ecology
    - Positive selection driving this at the level of circadian clock genes

→ modern working hypotheses discuss tradeoffs between energetic conservation in sleep and niche exploitation, describing it as an “energy allocation economy”

Lesku, J. A., & Schmidt, M. H. (2022). Energetic costs and benefits of sleep. Current biology : CB, 32(12), R656–R661. https://doi.org/10.1016/j.cub.2022.04.004

Extant questions, limitations

- Lack of quality sleep data for most species
- Currently, little insight into the evolutionary basis of sleep's fundamental role in human (+ other spp) health/behavior, poor synthesis of ecological theories and functional data
    - If sleep presents so many risks to fitness, what adaptive benefits makes it so ubiquitous, and how did sleep itself evolve?
    - Difficult to encompass all features of sleep: neurobiological/behavioral, thermoregulatory/metabolic, fitness/ecological into one unifying theory

Citations

Capellini, I., Barton, R. A., McNamara, P., Preston, B. T., & Nunn, C. L. (2008). Phylogenetic analysis of the ecology and evolution of mammalian sleep. Evolution; international journal of organic evolution, 62(7), 1764–1776. https://doi.org/10.1111/j.1558-5646.2008.00392.x

Lesku, J. A., & Schmidt, M. H. (2022). Energetic costs and benefits of sleep. Current biology : CB, 32(12), R656–R661. https://doi.org/10.1016/j.cub.2022.04.004

Lyamin, O. I., Kosenko, P. O., Korneva, S. M., Vyssotski, A. L., Mukhametov, L. M., & Siegel, J. M. (2018). Fur Seals Suppress REM Sleep for Very Long Periods without Subsequent Rebound. Current biology : CB, 28(12), 2000–2005.e2. https://doi.org/10.1016/j.cub.2018.05.022

Lyamin, O. I., & Siegel, J. M. (2019). Sleep in Aquatic Mammals. Handbook of behavioral neuroscience, 30, 375–393. https://doi.org/10.1016/b978-0-12-813743-7.00025-6

Yin, D., Zhong, Z., Zeng, F., Xu, Z., Li, J., Ren, W., Yang, G., Wang, H., & Xu, S. (2025). Evolution of canonical circadian clock genes underlies unique sleep strategies of marine mammals for secondary aquatic adaptation. PLoS genetics, 21(3), e1011598. https://doi.org/10.1371/journal.pgen.1011598